



KTU NOTES

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TIME DOMAIN ANALYSIS (MODULE-2)

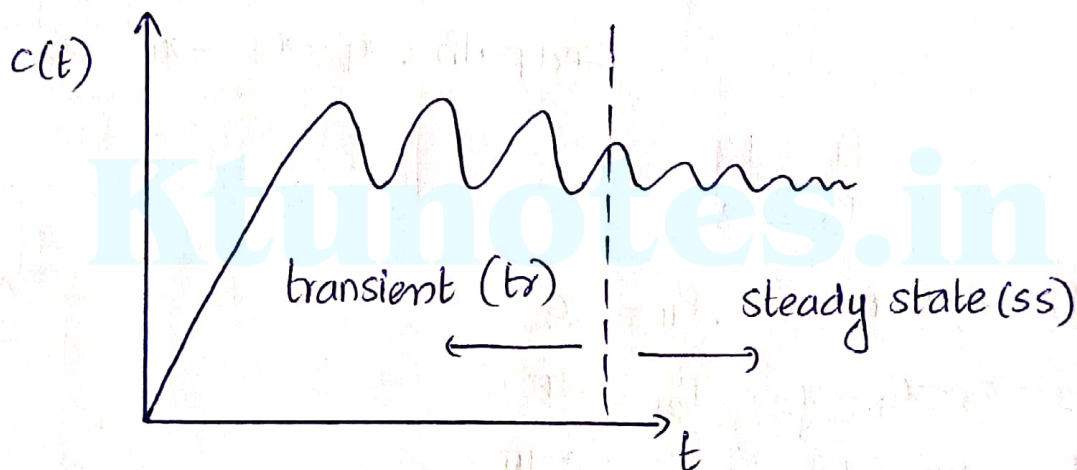
$g(t)$ = system
 $x(t)$ = i/p

$$c(t) = x(t) \otimes g(t)$$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s) \cdot H(s)}$$

$$C(s) = R(s) \cdot \frac{G(s)}{1 + G(s)H(s)}$$

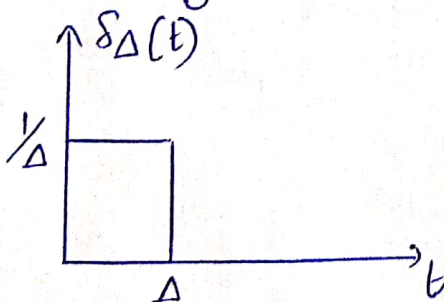
$$c(t) = \mathcal{L}^{-1}(C(s)) = \mathcal{L}^{-1} \left\{ \frac{G(s)}{1 + G(s)H(s)} R(s) \right\}$$



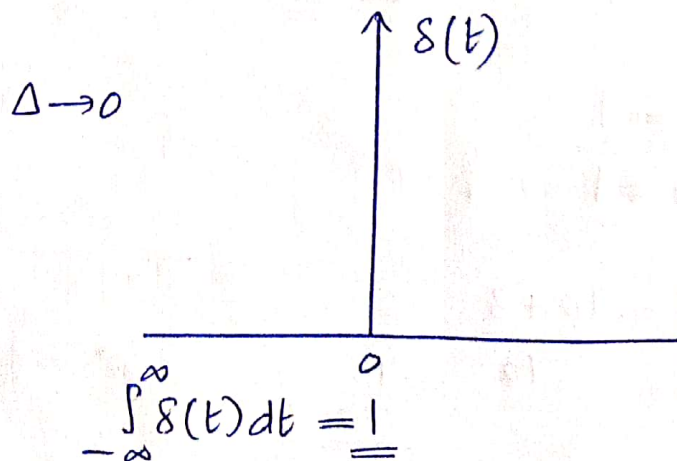
$$c(t) = c_{tr}(t) + c_{ss}(t)$$

Standard Test Signals

1. Impulse signal: ($\delta(t)$)



$$\mathcal{L}\{\delta(t)\} = 1$$

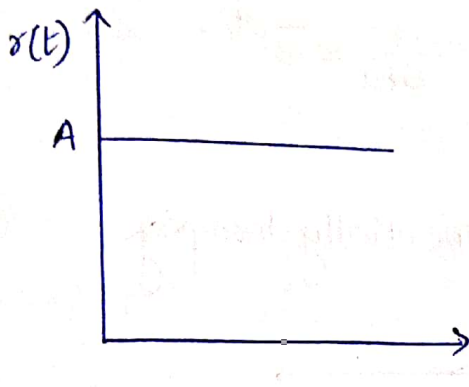


$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

$$\delta(t) \neq 0, t=0$$

$$= 0, t \neq 0$$

2. Step Signal



$$r(t) = Au(t)$$

↳ unit step signal

$$u(t) = 1, t \geq 0$$

$$= 0, t < 0$$

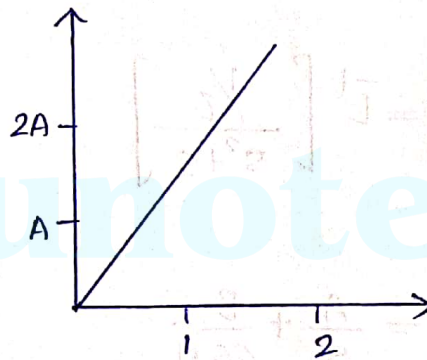
$$R(s) = L\{r(t)\} = \underline{\underline{\frac{A}{s}}}$$

$$L\{u(t)\} = \underline{\underline{\frac{1}{s}}}$$

3. Ramp Signal

$$r(t) = A \cdot t$$

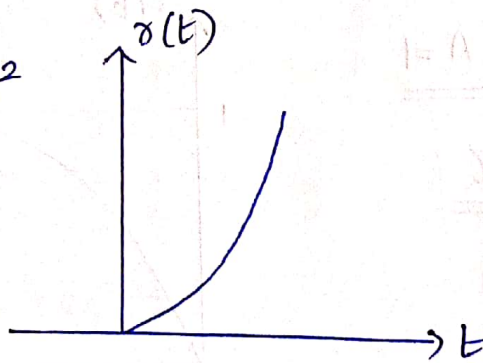
$$R(s) = \underline{\underline{\frac{A}{s^2}}}$$



4. Parabolic Signal

$$r(t) = \frac{A}{2} t^2$$

$$L\{r(t)\} = \frac{A}{s^3}$$



Order of a system

First order system

$$\tau \frac{dc(t)}{dt} + c(t) = r(t)$$

$$c(t) \text{ — o/p}$$

$$r(t) \text{ — i/p}$$

τ — Time constant of the system

$$TF \Rightarrow \frac{C(s)}{R(s)} = \frac{1}{\tau s + 1}$$

Impulse response of first order

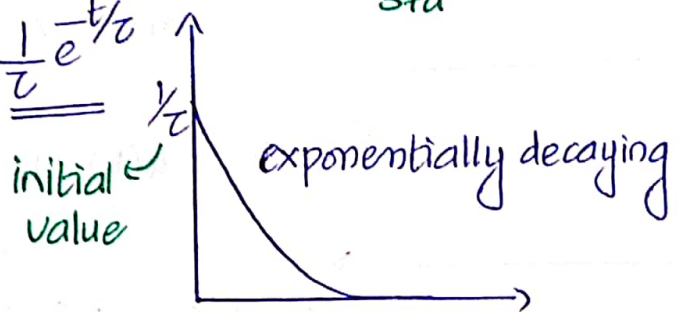
$$\text{I/p: } r(t) = \delta(t) \quad R(s) = 1$$

$$C(s) = R(s) \cdot \frac{1}{\tau s + 1} = \frac{1}{\tau s + 1}$$

$$C(t) = \mathcal{L}^{-1} \left\{ \frac{1}{\tau s + 1} \right\} = \frac{1/\tau}{s + 1/\tau} = \frac{1}{\tau} e^{-t/\tau}$$

$$C(\infty) = \lim_{t \rightarrow \infty} C(t) = 0 \text{ - final value}$$

$$\frac{1}{s+a} = e^{-at}$$



Unit step response

$$r(t) = u(t) \quad R(s) = 1/s$$

$$C(s) = \frac{1}{s} \cdot \frac{1}{\tau s + 1}$$

$$C(t) = \mathcal{L}^{-1} \left\{ \frac{1}{s(\tau s + 1)} \right\} = \mathcal{L}^{-1} \left[\frac{1/\tau}{s^2 + s} \right]$$

$$C(t) = \mathcal{L}^{-1} \left\{ \frac{1}{s(\tau s + 1)} \right\}$$

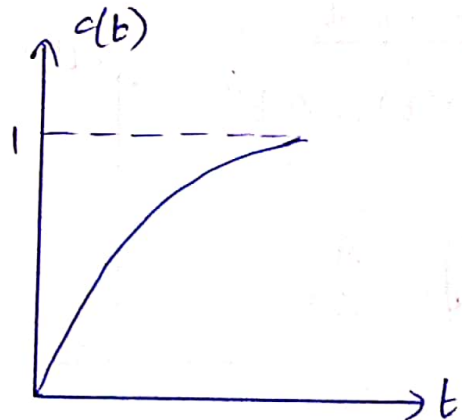
$$\frac{1}{s(\tau s + 1)} = \frac{A}{s} + \frac{B}{\tau s + 1}$$

$$A = \frac{1}{\tau s + 1} \Big|_{s=0} \Rightarrow A = 1$$

$$B = \frac{1}{s} \Big|_{s=-1/\tau} \Rightarrow B = -\tau$$

$$C(s) = \frac{1}{s} - \frac{\tau}{\tau s + 1}$$

$$C(t) = 1 - e^{-t/\tau}$$



$$t = \tau; C(t) = 1 - e^{-\tau/\tau} = 1 - e^{-1} = 0.63212$$

Time taken by the system to reach 63% of final steady state value $\rightarrow$ Time constant

$$C_1(t) = 1 - e^{-t/0.5}$$

$$C_2(t) = 1 - e^{-t/2}$$

$$C_3(t) = 1 - e^{-t/5}$$

Which is the fastest system?

$C_1(t)$ is fastest bcoz s/m reaches 63% of final state @ 0.5 seconds.

2nd order Systems

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

ω_n - Natural frequency of oscillation (rad/sec)

ζ = Damping ratio

$\zeta = 0$ - Undamped system

$0 < \zeta < 1$ - Underdamped system

$\zeta = 1$ - Critically damped system

$\zeta > 1$ - Overdamped system

Unit step response of undamped system:

$$x(t) = u(t)$$

$$R(s) = \frac{1}{s}$$

$$\zeta = 0$$

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + \omega_n^2}$$

$$C(s) = \frac{1}{s} \frac{\omega_n^2}{s^2 + \omega_n^2}$$

$$C(s) = \mathcal{L}^{-1} \left\{ \frac{\omega_n^2}{s(s^2 + \omega_n^2)} \right\}$$

$$\frac{\omega_n^2}{s(s^2 + \omega_n^2)} = \frac{A}{s} + \frac{Bs + C}{s^2 + \omega_n^2}$$

$$A = \frac{\omega_n^2}{s^2 + \omega_n^2} \bigg|_{s=0} = 1$$

$$\frac{\omega_n^2}{s(s^2 + \omega_n^2)} = \frac{1}{s} + \frac{Bs + C}{s^2 + \omega_n^2}$$

$$\frac{\omega_n^2}{s(s^2 + \omega_n^2)} = \frac{s^2 + \omega_n^2 + Bs^2 + Cs}{s(s^2 + \omega_n^2)}$$

$$\frac{(1+B)s^2 + Cs + \omega_n^2}{s(s^2 + \omega_n^2)}$$

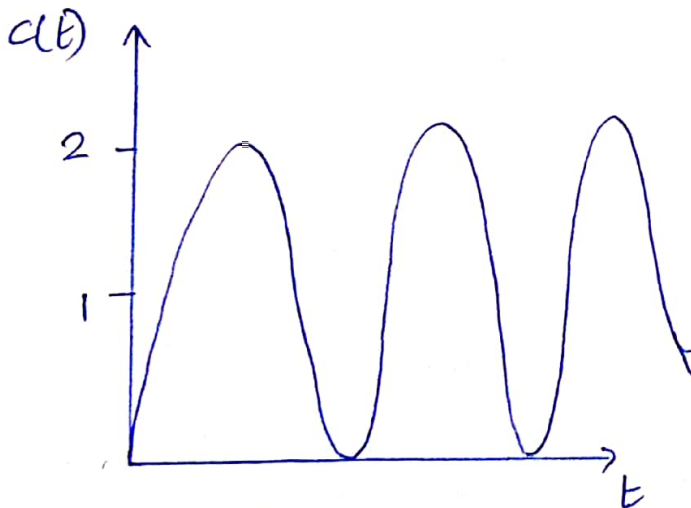
$$1+B=0, B=-1$$

$$C=0$$

$$\frac{\omega_n^2}{s(s^2 + \omega_n^2)} = \frac{1}{s} - \frac{s}{s^2 + \omega_n^2}$$

$$c(t) = \mathcal{L}^{-1} \left\{ \frac{1}{s} - \frac{s}{s^2 + \omega_n^2} \right\}$$

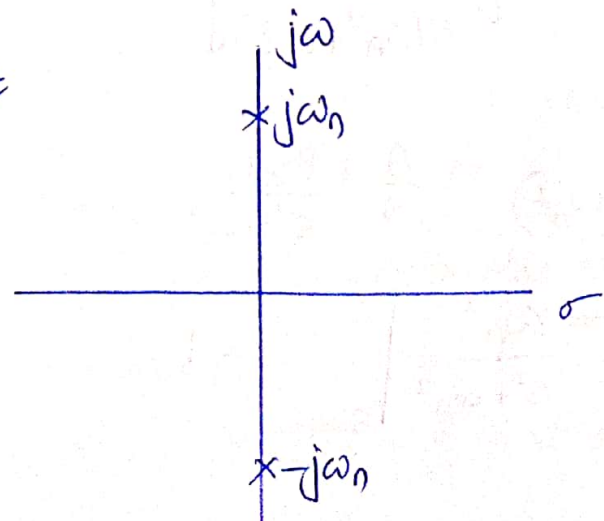
$$= \underline{\underline{1 - \cos \omega_n t}}$$



Poles

$$s^2 + \omega_n^2 = 0$$

$$s = \pm j\omega_n$$



Sustained oscillations

Freq of oscillation = ω_n rad/sec

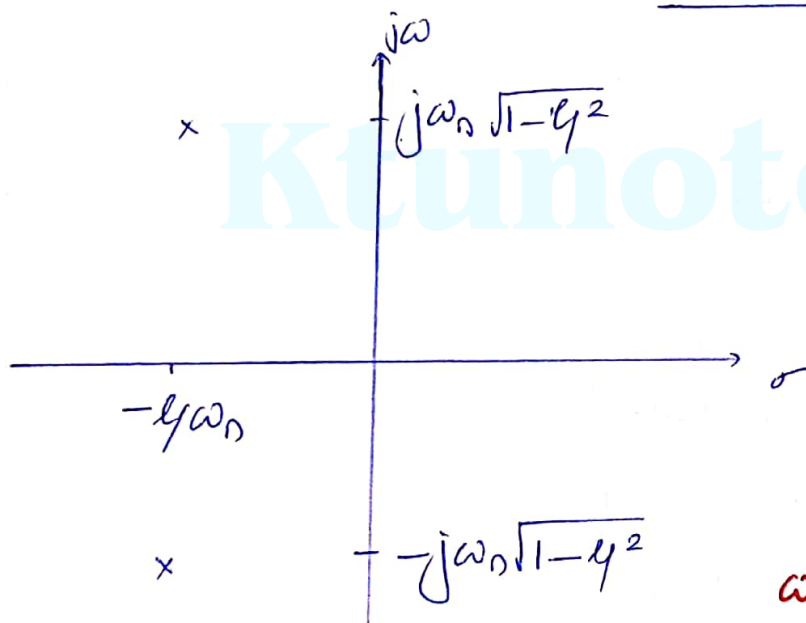
Unit step response of underdamped system

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$0 < \zeta < 1$$

Location of poles:

$$\begin{aligned} s &= \frac{-2\zeta\omega_n \pm \sqrt{4\zeta^2\omega_n^2 - 4\omega_n^2}}{2} \\ &= \frac{-2\zeta\omega_n \pm 2\omega_n\sqrt{\zeta^2 - 1}}{2} \\ &= -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} \\ &= \underline{\underline{-\zeta\omega_n \pm j\omega_n\sqrt{1 - \zeta^2}}} \end{aligned}$$



$$\omega_d = \omega_n\sqrt{1 - \zeta^2}$$

damped frequency of
oscillation (rad/sec)

Unit step response:

$$R(s) = \frac{1}{s}$$

$$C(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$A = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \bigg|_{s=0} = 1$$

$$\begin{aligned} \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} &= \frac{1}{s} + \frac{Bs + C}{s^2 + 2\zeta\omega_n s + \omega_n^2} \\ &= \frac{s^2 + 2\zeta\omega_n s + \omega_n^2 + Bs^2 + Cs}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} \\ &= \frac{(1+B)s^2 + (C + 2\zeta\omega_n)s + \omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} \end{aligned}$$

$$1+B=0$$

$$\Rightarrow B = -1$$

$$C + 2\zeta\omega_n = 0$$

$$\underline{\underline{C = -2\zeta\omega_n}}$$

$$C(s) = \mathcal{L}^{-1} \left\{ \frac{1}{s} + \frac{(-1)s - 2\zeta\omega_n}{s^2 + 2\zeta\omega_n s + \omega_n^2} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{1}{s} - \frac{(s + 2\zeta\omega_n)}{s^2 + 2\zeta\omega_n s + \omega_n^2} \right\}$$

$$C(s) = \frac{1}{s} - \frac{s + 2\zeta\omega_n}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$= \frac{1}{s} - \frac{s + \zeta\omega_n}{(s + \zeta\omega_n)^2 - \zeta^2\omega_n^2 + \omega_n^2} - \frac{\zeta\omega_n}{(s + \zeta\omega_n)^2 - \zeta^2\omega_n^2 + \omega_n^2}$$

$$= \frac{1}{s} - \frac{s + \zeta\omega_n}{(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)} - \frac{\zeta\omega_n}{(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)}$$

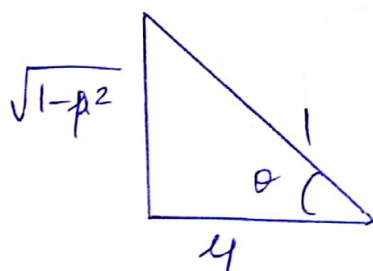
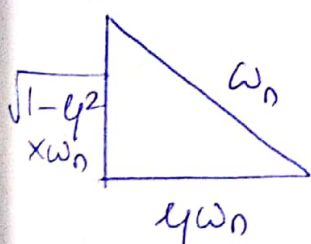
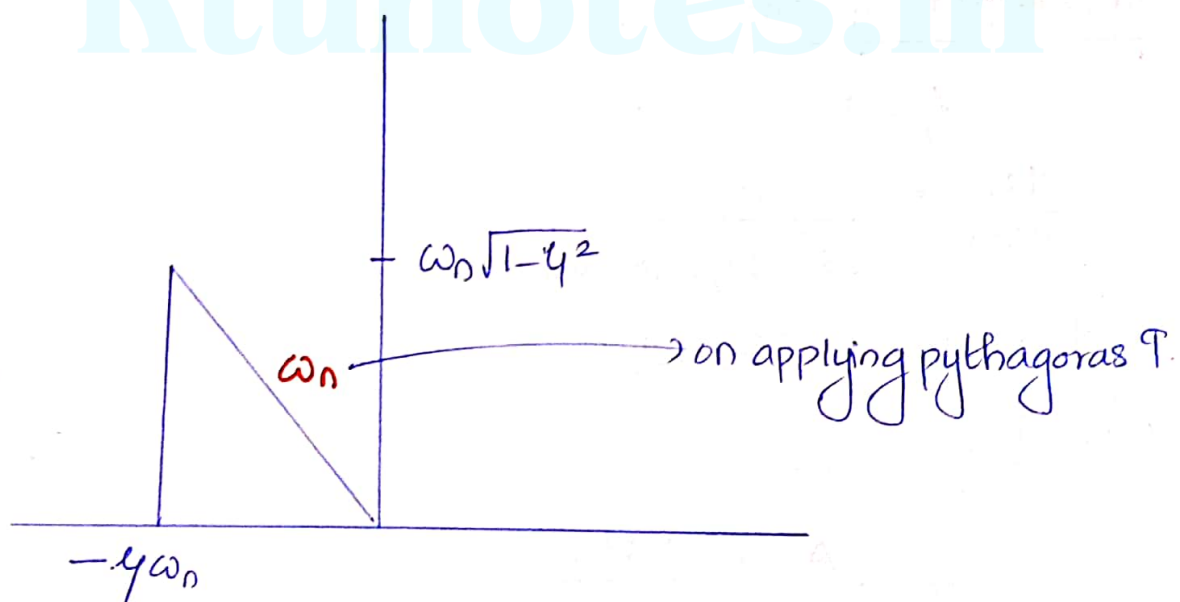
$$= \frac{1}{s} - \frac{s + \gamma \omega_n}{(s + \gamma \omega_n)^2 + \omega_d^2} - \frac{\gamma \omega_n}{(s + \gamma \omega_n)^2 + \omega_d^2} \times \frac{\omega_d}{\omega_d}$$

$$= \frac{1}{s} - \frac{s + \gamma \omega_n}{(s + \gamma \omega_n)^2 + \omega_d^2} - \frac{\gamma \omega_n \omega_d}{(s + \gamma \omega_n)^2 + \omega_d^2} \times \frac{1}{\omega_n \sqrt{1 - \gamma^2}}$$

$$= \frac{1}{s} - \frac{s + \gamma \omega_n}{(s + \gamma \omega_n)^2 + \omega_d^2} - \frac{\gamma \omega_d}{\sqrt{1 - \gamma^2} ((s + \gamma \omega_n)^2 + \omega_d^2)}$$

$$c(t) = 1 - e^{-\gamma \omega_n t} \cos \omega_d t - \frac{\gamma}{\sqrt{1 - \gamma^2}} e^{-\gamma \omega_n t} \sin \omega_d t$$

$$= 1 - \frac{e^{-\gamma \omega_n t}}{\sqrt{1 - \gamma^2}} \left[\sqrt{1 - \gamma^2} \cos \omega_d t + \gamma \sin \omega_d t \right]$$



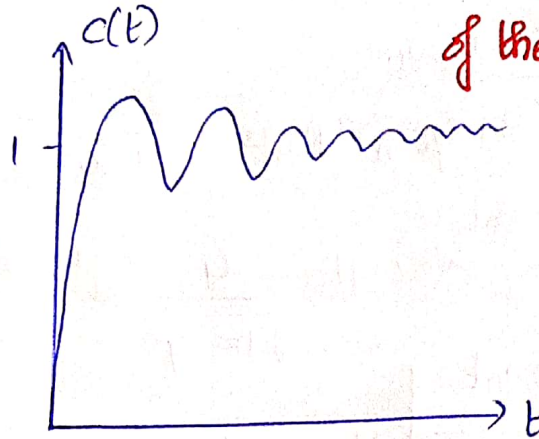
$$\sin \theta = \sqrt{1 - \gamma^2}$$

$$\cos \theta = \gamma$$

$$c(t) = 1 - \frac{e^{-\gamma \omega_n t}}{\sqrt{1-\gamma^2}} \left[\sin \theta \cos \omega_d t + \cos \theta \cdot \sin \omega_d t \right]$$

$$= 1 - \frac{e^{-\gamma \omega_n t}}{\sqrt{1-\gamma^2}} \sin(\omega_d t + \theta)$$

$c(\infty) = 1$
which is the
steady state value
of the system.



$$\theta = \tan^{-1} \left(\frac{\sqrt{1-\gamma^2}}{\gamma} \right)$$

Critically damped System

$$\gamma = 1$$

$$\frac{c(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\gamma\omega_n s + \omega_n^2}$$

For unit step response $R(s) = \frac{1}{s}$

$$c(s) = \frac{1}{s} \frac{\omega_n^2}{s^2 + 2\omega_n s + \omega_n^2}$$

$$\frac{\omega_n^2}{s(s^2 + 2\omega_n s + \omega_n^2)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 2\omega_n s + \omega_n^2}$$

$$A = \frac{\omega_n^2}{(s^2 + \omega_n^2)} \Big|_{s=0} = 1$$

$$\begin{aligned} \frac{\omega_n^2}{s(s^2 + 2\omega_n s + \omega_n^2)} &= \frac{1}{s} + \frac{Bs + C}{s^2 + 2\omega_n s + \omega_n^2} \\ &= \frac{Bs^2 + Cs + s^2 + 2\omega_n s + \omega_n^2}{s(s^2 + 2\omega_n s + \omega_n^2)} \end{aligned}$$

$$(1+B)=0$$

$$\Rightarrow B = -1$$

$$C = \alpha - 2\omega_n$$

$$C(s) = \frac{1}{s} + \frac{(-1)s(-2\omega_n)(-s-2\omega_n)}{(s+\omega_n)^2}$$

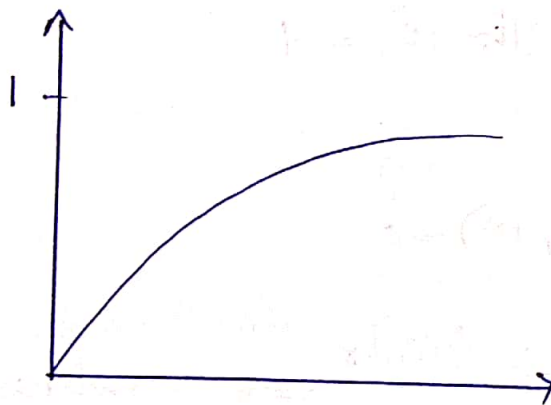
$$C(t) = \mathcal{L}^{-1} \left\{ \frac{1}{s} - \frac{s}{(s+\omega_n)^2} \right\}$$

$$C(s) = \frac{1}{s} - \frac{(s+2\omega_n)}{(s+\omega_n)^2}$$

$$= \frac{1}{s} - \frac{1}{s+\omega_n} - \frac{\omega_n}{(s+\omega_n)^2}$$

$$C(t) = 1 - e^{-\omega_n t} - e^{-\omega_n t} \cdot t \cdot \omega_n$$

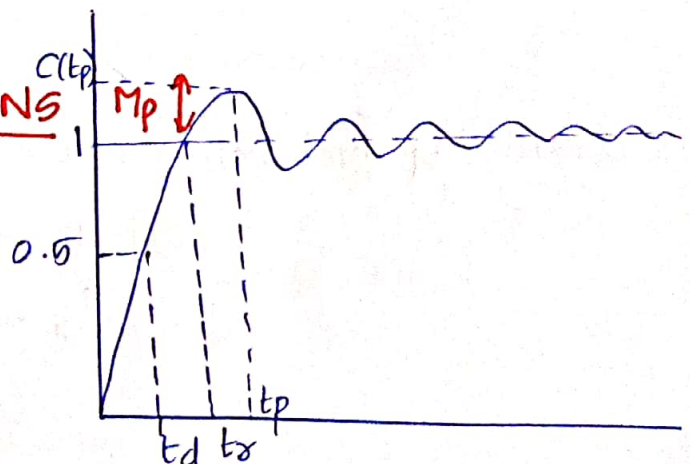
$$= 1 - e^{-\omega_n t} (1 + \omega_n t)$$



HW) Overdamped system

TYPICAL TIME DOMAIN SPECIFICATIONS

Delay time (t_d) - Time taken for the response to reach 50% of final value.



- Rise time (t_r): Time taken by the response to reach the final value for the very first time.
- Peak time (t_p): Time taken for the response to reach the peak time.

- Peak overshoot (M_p): $M_p = \frac{c(t_p) - c(\infty)}{c(\infty)}$

$$\% ; M_p = \frac{c(t_p) - c(\infty)}{c(\infty)} \times 100\%$$

- Setting time (t_s): Time taken to reach within 2% or 5% tolerance level.

Rise time (t_r)

$$c(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \theta)$$

At $t = t_r$

$$c(t_r) = 1$$

$$1 - \frac{e^{-\zeta\omega_n t_r}}{\sqrt{1-\zeta^2}} \sin(\omega_d t_r + \theta) = 1$$

$$\frac{e^{-\zeta\omega_n t_r}}{\sqrt{1-\zeta^2}} \sin(\omega_d t_r + \theta) = 0$$

$\frac{e^{-\zeta\omega_n t_r}}{\sqrt{1-\zeta^2}} \neq 0 \therefore$ we have a definite risetime

$$\sin(\omega_d t_r + \theta) = 0$$

$$\omega_d t_r + \theta = 0, \pi, 2\pi, 3\pi, \dots$$

Considering first non-zero value,

$$\omega_d t_r + \theta = \pi$$

$$t_r = \frac{\pi - \theta}{\omega_d}$$

$$\theta = \tan^{-1} \left(\frac{\sqrt{1-\zeta^2}}{\zeta} \right)$$

$$\omega_d = \omega_n \sqrt{1-\zeta^2}$$

$$t_p = \frac{\pi - \tan^{-1} \left(\frac{\sqrt{1-\zeta^2}}{\zeta} \right)}{\omega_n \sqrt{1-\zeta^2}}$$

Peak time (t_p)



$$\frac{d}{dt} c(t) = 0 \text{ at } t = t_p$$

$$\frac{d}{dt} c(t) = \frac{d}{dt} \left[1 - \frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \theta) \right]$$

$$= -\frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} (-\zeta \omega_n) \sin(\omega_d t + \theta) - \frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} \omega_n \cos(\omega_d t + \theta)$$

$$= \frac{\omega_n e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} \left[\zeta \sin(\omega_d t + \theta) - \sqrt{1-\zeta^2} \cos(\omega_d t + \theta) \right]$$

$$= \frac{\omega_n e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} \left[\cos \theta \cdot \sin(\omega_d t + \theta) - \sin \theta \cdot \cos(\omega_d t + \theta) \right]$$

$$\frac{\omega_n e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \theta - \theta)$$

$$= \frac{\omega_n e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} \sin \omega_d t$$

At $t = t_p$,

$$\frac{d}{dt} c(t) = 0$$

$$\frac{d}{dt} (c(t_p)) = 0 \Rightarrow \frac{\omega_n e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} \sin \omega_d t_p = 0$$

$$\sin \omega_d t_p = 0$$

$$\omega_d t_p = 0, \pi, \dots$$

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$$

Peak Overshoot

$$M_p = \frac{c(t_p) - c(\infty)}{c(\infty)}$$

$$c(t_p) = 1 - \frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \theta)$$

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$$

$$c(t_p) = 1 - \frac{e^{-\frac{\zeta \pi}{\sqrt{1-\zeta^2}}}}{\sqrt{1-\zeta^2}} (\sin \pi + \theta)$$

$$= 1 + \frac{e^{-\frac{4\pi}{\sqrt{1-\zeta^2}}} \sin \theta}{\sqrt{1-\zeta^2}}$$

$$= 1 + e^{-4\pi/\sqrt{1-\zeta^2}}$$

$$M_p = \frac{1 + e^{-4\pi/\sqrt{1-\zeta^2}} - 1}{1}$$

$$= e^{-4\pi/\sqrt{1-\zeta^2}}$$

Settling time (t_s)

$$c(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \theta)$$

$$2\% : \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} = 0.02$$

For small values of ζ , $\sqrt{1-\zeta^2} \approx 1$,

$$e^{-\zeta\omega_n t_s} = 0.02$$

$$t_s = \frac{4}{\zeta\omega_n}$$

$$5\% : \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} = 0.05$$

For small values of ζ , $\sqrt{1-\zeta^2} \approx 1$

$$e^{-\zeta\omega_n t_s} = 0.05$$

$$t_s = \frac{3}{\zeta\omega_n}$$

Q: Find the time domain specifications of following s/m where i/p and o/p are related as:

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 8y(t) = 8x(t)$$

Laplace trans: $s^2Y(s) + 4sY(s) + 8Y(s) = 8X(s)$

$$\frac{Y(s)}{X(s)} = \frac{8}{s^2 + 4s + 8} \quad \text{Transfer fn}$$

$$\omega_n = 2.828 \text{ rad/s}$$

$$\zeta = 0.707$$

It is an underdamped system

$$t_r = \frac{\pi - \tan^{-1}\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right)}{\omega_n \sqrt{1-\zeta^2}}$$

$$= \frac{\pi - \tan^{-1}\left(\frac{\sqrt{1-(0.707)^2}}{0.707}\right)}{2.828 \sqrt{1-(0.707)^2}}$$

$$= \underline{\underline{1.17 \text{ sec}}}$$

$$t_p = \frac{\pi}{\omega_d} = \underline{\underline{1.57 \text{ sec}}}$$

$$\% M_p = e^{-\zeta\pi/\sqrt{1-\zeta^2}} \times 100\%$$

$$= e^{-0.707 \times 3.14 / \sqrt{1-(0.707)^2}} \times 100\%$$

$$= \underline{\underline{4.32\%}}$$

2% setting time

$$t_s = \frac{4}{\gamma \omega_n} = \underline{\underline{2 \text{ secs}}}$$

5% setting time

$$t_s = \frac{3}{\gamma \omega_n} = \underline{\underline{1.5 \text{ secs}}}$$

Q: The response of a dm is found to be $c(t) = 1 + 0.2e^{-60t} - 1.2e^{-10t}$ when it is subject to unit step i/p. Determine the natural freq of oscillation and damping ratio.

$$c(t) = 1 + 0.2e^{-60t} - 1.2e^{-10t}$$

$$c(s) = \frac{1}{s} + \frac{0.2}{s+60} - \frac{1.2}{s+10}$$

$$C(s) = R(s) \cdot (TF)$$

$$R(s) \cdot TF = \frac{1}{s} + \frac{0.2}{s+60} - \frac{1.2}{s+10}$$

$$\frac{1}{s} TF = \frac{1}{s} + \frac{0.2}{s+60} - \frac{1.2}{s+10}$$

$$TF = 1 + \frac{0.2s}{s+60} - \frac{1.2s}{s+10} = \underline{\underline{\frac{600}{s^2 + 70s + 600}}}$$

$$\omega_n = 24.5 \text{ rad/sec}$$

$$\gamma = \underline{\underline{1.428}}$$

Qn: The unit impulse response of a system is $-4e^{-t} + 6e^{-2t}$. Find the unit step response of the system.

$$g(t) = -4e^{-t} + 6e^{-2t}$$

$$G(s) = \frac{-4}{s+1} + \frac{6}{s+2}$$

$$\frac{C(s)}{R(s)} = G(s)$$

for unit step inp, $R(s) = \frac{1}{s}$

$$C(s) = R(s) \cdot G(s)$$

$$= \frac{1}{s} \left(\frac{-4}{s+1} + \frac{6}{s+2} \right)$$

$$= \frac{-4s - 8 + 6s + 6}{s(s+1)(s+2)}$$

$$= \frac{2s - 2}{s(s+1)(s+2)}$$

$$\frac{2s - 2}{s(s+1)(s+2)} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+2}$$

$$A = \left. \frac{2s - 2}{(s+1)(s+2)} \right|_{s=0} = \frac{-2}{2} = \underline{\underline{-1}}$$

$$B = \left. \frac{2s - 2}{s(s+2)} \right|_{s=-1} = \frac{-4}{-1} = \underline{\underline{4}}$$

$$C = \left. \frac{2s - 2}{s(s+1)} \right|_{s=-2} = \frac{-6}{2} = \underline{\underline{-3}}$$

$$c(s) = \frac{-1}{s} + \frac{4}{s+1} - \frac{3}{s+2}$$

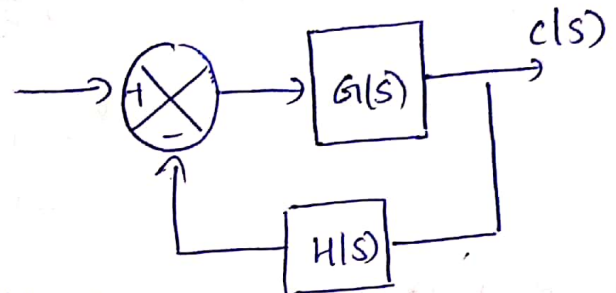
$$= -1 + 4e^{-t} - 3e^{-2t}$$

Qn: The unity feedback s/m is characterised by an open loop transfer function

$$G(s) = \frac{k}{s(s+10)}$$

Determine the ^{gain k} decay so that the s/m will have a

damping factor of 0.5. Also determine settling time, peak time, peak, overshoot, for unit step i/p



$$\text{Open loop TF} = G(s) \cdot H(s)$$

here, $\therefore$ it is unity feedback;

$$H(s) = 1$$

$$\therefore \text{TF} = G(s) \quad \{\text{open loop}\} \quad \text{--- (1)}$$

closed loop transfer function:

$$\text{CL TF} = \frac{G(s)}{1 + G(s)H(s)}$$

$$\text{TF} = \frac{G(s)}{1 + G(s)} \quad \text{--- (2)} \quad \{\text{Closed loop}\}$$

$$= \frac{k}{s(s+10)} = \frac{k}{s^2 + 10s + k}$$

$$1 + \frac{k}{s(s+10)}$$

$$= \frac{k}{s^2 + 10s + k}$$

$$\text{2nd order s/m TF} = \frac{\omega_n^2}{s^2 + 2\gamma\omega_n s + \omega_n^2}$$

$$\omega_n^2 = k$$

$$2\gamma\omega_n = 10$$

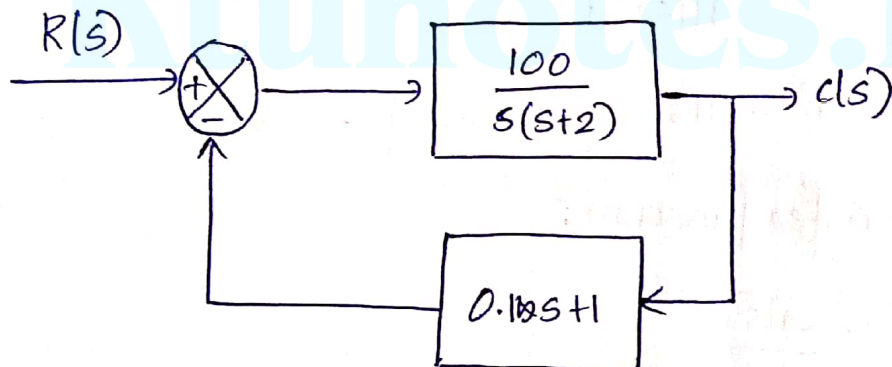
$$\gamma = \frac{10}{2\sqrt{k}}$$

$$\text{Given, } \gamma = 0.5$$

$$\Rightarrow 0.5 = \frac{10}{2\sqrt{k}}$$

$$k = 10^2 \\ = 100$$

Qn: Find the unit step response of the system



For unit step response, $R(s) = 1$

$$\begin{aligned} \text{TF} &= \frac{\frac{100}{s(s+2)}}{1 + \frac{100}{s(s+2)} (0.1s+1)} = \frac{100}{s^2 + 2s + 10s + 100} \\ &= \frac{100}{s^2 + 12s + 100} \end{aligned}$$

$$\text{2nd order s/m TF} = \frac{\omega_n^2}{s^2 + 2\gamma\omega_n s + \omega_n^2}$$

$$\omega_n = 10$$

$$2\gamma \times 10 = 12$$

$$\gamma = 0.6$$

So, it is an underdamped system

$$C(s) = R(s) \cdot TF$$

$$= \frac{1}{s} \left(\frac{100}{s^2 + 12s + 100} \right)$$

$$= \frac{A}{s} + \frac{Bs + C}{s^2 + 12s + 100}$$

$$A = \left. \frac{100}{s^2 + 12s + 100} \right|_{s=0} = 1$$

$$\frac{100}{s(s^2 + 12s + 100)} = \frac{1}{s} + \frac{Bs + C}{s^2 + 12s + 100}$$

$$= \frac{s^2 + 12s + 100 + Bs^2 + Cs}{s(s^2 + 12s + 100)}$$

$$(B + 1) = 0$$

$$\underline{\underline{B = -1}}$$

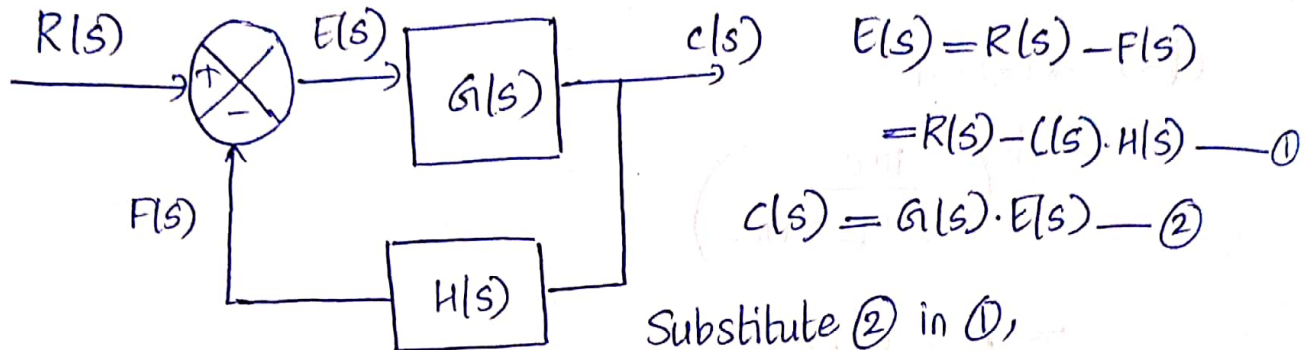
$$\underline{\underline{C = -12}}$$

$$C(s) = \frac{1}{s} - \frac{(s + 12)}{s^2 + 12s + 100}$$

$$= \frac{1}{s} - \frac{(s + 12)}{(s + 6)^2 + 8^2} = \frac{1}{s} - \frac{(s + 6)}{(s + 6)^2 + 8^2} - \frac{6 \times 8}{8[(s + 6)^2 + 8^2]}$$

$$= 1 - e^{-6t} \cos 8t - \frac{6}{8} e^{-6t} \sin 8t$$

Steady State error



Substitute (2) in (1),

$$E(s) = R(s) - G(s) E(s) H(s)$$

$$E(s) [1 + G(s) H(s)] = R(s)$$

$$E(s) = \frac{R(s)}{1 + G(s) H(s)}$$

$\rightarrow$ ilp of the s/m
open loop TF

$G(s) H(s)$ - No of poles of OL TF lying at origin.

$$e(t) = \mathcal{L}^{-1}\{E(s)\} = \mathcal{L}^{-1}\left\{\frac{R(s)}{1 + G(s) H(s)}\right\}$$

Steady state error, $e_{ss} = \lim_{t \rightarrow \infty} e(t)$.

$$= \lim_{s \rightarrow 0} s E(s)$$

Static error constants

① Positional error constant $K_p = \lim_{s \rightarrow 0} G(s) \cdot H(s)$

② Velocity error constant $K_v = \lim_{s \rightarrow 0} s \cdot G(s) H(s)$

③ Acceleration error constant $k_a = \lim_{s \rightarrow 0} s^2 \cdot G(s)H(s)$

Type number of the system

$$G(s)H(s) = \frac{k(s+z_1)(s+z_2)(s+z_3) \dots}{s^N (s+p_1)(s+p_2)(s+p_3) \dots}$$

at origin (N poles @ origin)

N=0 - Type 0 system

N=1 - Type 1 system

N=2 - Type 2 system

Steady state error when it is unit step signal:

$$\begin{aligned} e_{ss} &= \lim_{s \rightarrow 0} \frac{SR(s)}{1 + G(s)H(s)} \\ &= \lim_{s \rightarrow 0} \frac{s \cdot 1/s}{1 + G(s)H(s)} \\ &= \frac{1}{1 + \lim_{s \rightarrow 0} G(s)H(s)} \\ &= \frac{1}{1 + K_p} \end{aligned}$$

Type - 0 s/m

$$G(s)H(s) = \frac{k(s+z_1)(s+z_2) \dots}{(s+p_1)(s+p_2) \dots}$$

$$\begin{aligned}
 K_p &= \lim_{s \rightarrow 0} G(s)H(s) \\
 &= \frac{K \bar{\tau}_1 \bar{\tau}_2 \bar{\tau}_3 \dots}{P_1 P_2 P_3 \dots} \\
 &= \underline{\underline{\text{Constant}}}
 \end{aligned}$$

$$\begin{aligned}
 \text{Steady state error} &= \frac{1}{1+K_p} \\
 &= \underline{\underline{\text{Constant}}}
 \end{aligned}$$

Type - 1 s/m

$$\begin{aligned}
 G(s)H(s) &= \frac{K(s+\bar{\tau}_1)(s+\bar{\tau}_2)(s+\bar{\tau}_3) \dots}{s^N(s+P_1)(s+P_2)(s+P_3) \dots} \\
 &= \frac{K(0+\bar{\tau}_1)(0+\bar{\tau}_2)(0+\bar{\tau}_3) \dots}{s(0+P_1)(0+P_2)(0+P_3) \dots} \\
 &= \frac{\infty}{\infty}
 \end{aligned}$$

put $s=0$

$$e_{ss} = \frac{1}{1+\infty} = \underline{\underline{0}}$$

Type - 2 s/m

$$\begin{aligned}
 G(s)H(s) &= \frac{K(\bar{\tau}_1 \bar{\tau}_2 \bar{\tau}_3 \dots)}{s^2 P_1 P_2 P_3 \dots} \\
 &= \underline{\underline{\infty}}
 \end{aligned}$$

$$e_{ss} = 0$$

Steady state error when i/p is unit ramp

$$R(s) = 1/s^2$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{SR(s)}{1+G(s)H(s)} = \lim_{s \rightarrow 0} \frac{1}{s+G(s)H(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{G(s)H(s)}$$

$$= \frac{1}{K_v}$$

Type-0 s/m

$$G(s)H(s) = \frac{k z_1 z_2 z_3 \dots}{P_1 P_2 P_3 \dots}$$

$$e_{ss} = K_v = \lim_{s \rightarrow 0} s G(s)H(s)$$

$$= \lim_{s \rightarrow 0} \frac{s k z_1 z_2 z_3}{P_1 P_2 P_3}$$

$$= \underline{\underline{0}}$$

$$e_{ss} = \underline{\underline{\infty}}$$

Type-1 s/m

$$K_v = \infty \text{ constant}$$

$$e_{ss} = \text{constant}$$

Type-2 s/m

$$K_v = \infty$$

$$e_{ss} = 0$$

Parabolic i/p

$$R(s) = \frac{1}{s^3}$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s R(s)}{1 + G(s)H(s)}$$

$$= \lim_{s \rightarrow 0} \frac{R(s)}{s^2 + s^2 G(s)H(s)}$$

$$= \frac{1}{\underline{\underline{K_a}}}$$

Type-0 s/m

$$s^2 G(s)H(s) = K_a = 0$$

$$e_{ss} = \infty$$

Type-1 s/m

$$K_a = 0$$

$$e_{ss} = \infty$$

Type-2 s/m

$$K_a = \text{constant}$$

$$e_{ss} = \text{constant}$$

Type-3 s/m

$$K_a = \infty$$

$$e_{ss} = 0$$

Type No.	Unit Step	Ramp	Parabolic
0	$\frac{1}{1+k_p}$	∞	∞
1	0	$\frac{1}{k_v}$	∞
2	0	0	$\frac{1}{k_a}$
3	0	0	0

a) For a unity feedback control s/m, the open loop TF $G(s) = \frac{10(s+2)}{s^2(s+1)}$. Find
a: the static error constants

b: the steady state error when i/p $R(s) = \frac{3}{s} - \frac{2}{s^2} + \frac{1}{3s^3}$.

$$K_p = \lim_{s \rightarrow 0} G(s) \cdot H(s)$$

$$K_v = \lim_{s \rightarrow 0} s \cdot G(s) \cdot H(s)$$

$$K_a = \lim_{s \rightarrow 0} s^2 G(s) H(s)$$

Here, $H(s) = 1$

$$K_p = \lim_{s \rightarrow 0} \frac{10(s+2)}{s^2(s+1)} = \frac{20}{0} = \infty$$

$$\begin{aligned} K_v &= \lim_{s \rightarrow 0} s G(s) H(s) = \lim_{s \rightarrow 0} s \times \frac{10(s+2)}{s^2(s+1)} \\ &= \lim_{s \rightarrow 0} \frac{10s+20}{s^2+s} = \infty \end{aligned}$$

$$K_a = \lim_{s \rightarrow 0} s^2 G(s) H(s) = \lim_{s \rightarrow 0} \frac{10s+20}{s+1} = \underline{\underline{20}}$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s \cdot R(s)}{1 + G(s)H(s)}$$

$$= \lim_{s \rightarrow 0} \frac{s \left(\frac{3}{s} - \frac{2}{s^2} + \frac{1}{3s^3} \right)}{1 + \frac{10(s+2)}{s^2(s+1)}}$$

$$= \lim_{s \rightarrow 0} \frac{\left(3 - \frac{2}{s} + \frac{1}{3s^2} \right) (s^3 + s^2)}{s^3 + s^2 + 10s + 20}$$

$$= \frac{(9s^2 - 6s + 1)(s^3 + s^2)}{3s^2}$$

$$= \frac{(9s^2 - 6s + 1)(s^3 + s^2)}{3s^2(s^3 + s^2 + 10s + 20)}$$

$$= \frac{(9s^2 - 6s + 1)(s + 1)}{3(s^3 + s^2 + 10s + 20)}$$

$$= \underline{\underline{\frac{1}{60}}}$$

Q) For an open loop TF with unity FB, $G(s) = \frac{k(s+5)}{s(s+6)(s+7)(s+9)}$

Find the ilp to the system and the value of k so that there is 5% error in steady state.

It is a type 1 s/m.

5% error is present

$\Rightarrow k$ is a constant

So $R(s)$ is a ramp i/p

$$e_{ss} = 0.05$$

$$0.05 = \lim_{s \rightarrow 0} \frac{1}{K_v}$$

$$K_v = \lim_{s \rightarrow 0} s G(s) H(s)$$

$$= \lim_{s \rightarrow 0} \frac{s k (s+5)}{s(s+6)(s+7)(s+9)} \cdot 1$$

$$\lim_{s \rightarrow 0} \frac{k (s+5)}{(s+6)(s+7)(s+9)}$$

$$\frac{1}{0.05} = \frac{5k}{378}$$

$$0.05 = \frac{5k}{378}$$

$$k = \frac{0.05 \times 378}{5 \times 0.05}$$

$$\underline{\underline{k = 378}}$$

$$\underline{\underline{= 378 - 1512}}$$

$$e_{ss} =$$

Q) For the following s/m with unity FB, explain what type of i/p signal give rise to constant steady state error. Also calculate steady state error value.

$$G(s) = \frac{10}{(s+2)(s+3)}$$

Type 0 $\Rightarrow$ i/p is unit step.

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s \times \frac{1}{s}}{1 + \frac{10}{(s+2)(s+3)}}$$

$$= \lim_{s \rightarrow 0} \frac{1(s+2)(s+3)}{s^2 + 6s + 10}$$

$$= \frac{s^2 + 5s + 6}{s^2 + 6s + 10}$$

$$= \frac{6}{10}$$

$$= \underline{\underline{\frac{3}{5}}}$$

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